Electric Vehicle Price Prediction in the Vietnamese Market

A Web-Scraped Dataset with LLM-Assisted Feature Extraction

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Outline

1. Introduction
2. Data Collection
3. Data Quality Analysis
4. Feature Engineering
5. Experimental Setup
6. Results
7. Discussion & Conclusion

Motivation

- Vietnam's EV market is **growing rapidly**
  - VinFast mass-producing since 2022
  - Chinese imports (BYD, Wuling, Geely)
  - Government ICE phase-out by 2040
- **No structured dataset** exists for Vietnamese EV prices
- No local equivalent of Kelley Blue Book or AutoTrader
- Listings are **unstructured, Vietnamese-language**, scattered across multiple platforms

Gap

Existing car price prediction studies rely on clean, single-source, Western-market datasets

Our Goal

End-to-end pipeline: **scraping** $\rightarrow$ **LLM extraction** $\rightarrow$ **data quality** $\rightarrow$ **prediction**

Contributions

1. **First Vietnamese EV dataset** — 4,166 records from 4 marketplaces with unified 20-column schema
2. **Two-stage LLM extraction** — local Qwen 2.5 3B + cloud GPT-5-nano with auto-retry and Pydantic validation
3. **Data quality analysis** — identified and corrected 3 critical issues, reducing RMSE by **67%** (1.02B $\rightarrow$ 332M VND)
4. **Segment-specific evaluation** — practical accuracy (RMSE $<$ 100M) for budget & mid-range segments (92% of test records)

Key Finding

Data quality remediation contributes **far more** to accuracy than model selection (67% vs 15% RMSE improvement).

Data Sources

Website	Method	Records
bonbanh.com	BeautifulSoup	1,493
chotot.com	Scrapy	1,666
xevinfastluot.com	Selenium	279
otodien.vn	BeautifulSoup	728
Total		**4,166**

Unified schema: 20 columns covering ID, specs, condition, metadata, and price

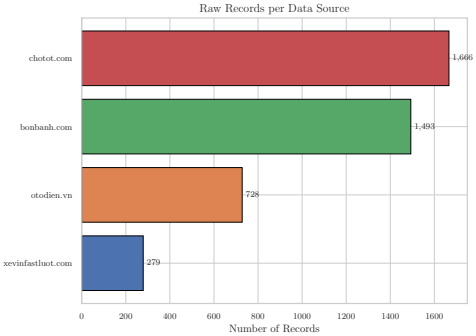


Figure: Records per source

LLM-Assisted Feature Extraction

Two-Stage Pipeline

1. **Local — Qwen 2.5 3B** (via Ollama)
   - Zero-shot prompt → JSON output
   - Pydantic schema validation
   - Auto-retry up to 3 attempts on failure
2. **Cloud — GPT-5-nano** (verification)
   - Batch-processes low-confidence records
   - Corrects hallucinated trim names

Fill rates:

Brand: 99.3% Base model: 98.7%
Year: 98.3% Price: 100%

Failure mode:

Trim-level hallucination for non-VinFast brands

Three Critical Issues

1. chotot.com 10× Inflation

- 1,666 records (40%)
- chotot.com prices stored in 1,000 VND units
- Fix: divide by 10

2. BYD Pricing Error

- 90 records (2.3%)
- ∼56% of total weighted error
- Fix: remove all BYD

3. Duplicate Leakage

- 887 duplicates (22%)
- Inflated R^2: 0.787
- Honest R^2: 0.725

Stage	Records	Best RMSE
Raw (all issues)	3,992	1.02B VND
+ All fixes applied	3,015	**332M VND**

↓ **67% RMSE reduction**

chotot.com Price Inflation

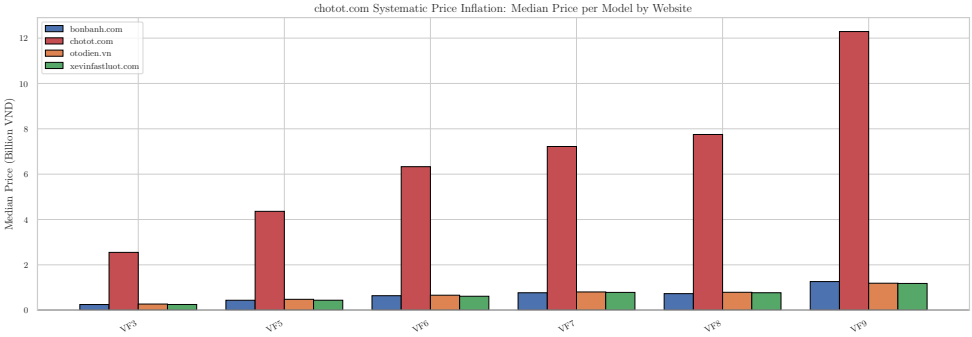


Figure: Systematic 10× price inflation from chotot.com vs other sources

Before vs After Data Quality Fixes

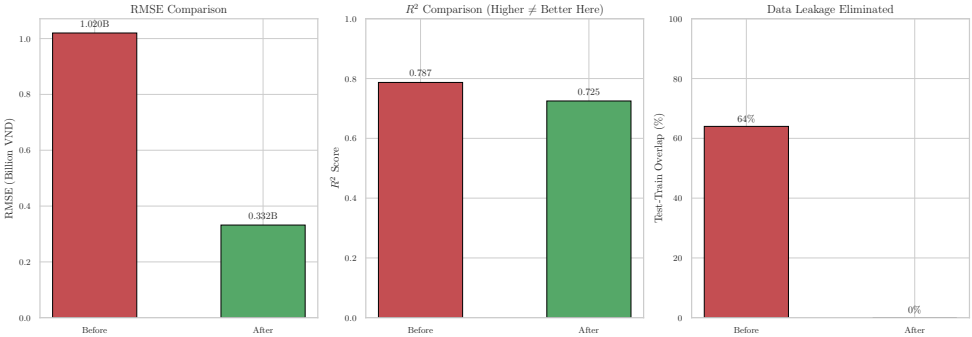


Figure: Model performance before and after corrections — "before" metrics were inflated by leakage

Feature Engineering Pipeline

41 features from 3,015 records:

Category	Count
Numeric (age, mileage, ...)	4
EV Specs (kWh, range, hp)	3
Target-encoded	2
One-hot (brand, model, ...)	32
Total	**41**

Key Decisions

- **EV specs recovery**: battery, range, power from otodien.vn lookup (29 models)
- **Hybrid encoding**: target-encode only high-cardinality features
- Target-encoding everything $\Rightarrow$ R^2 drops to 0.087!

EV Specifications Recovery

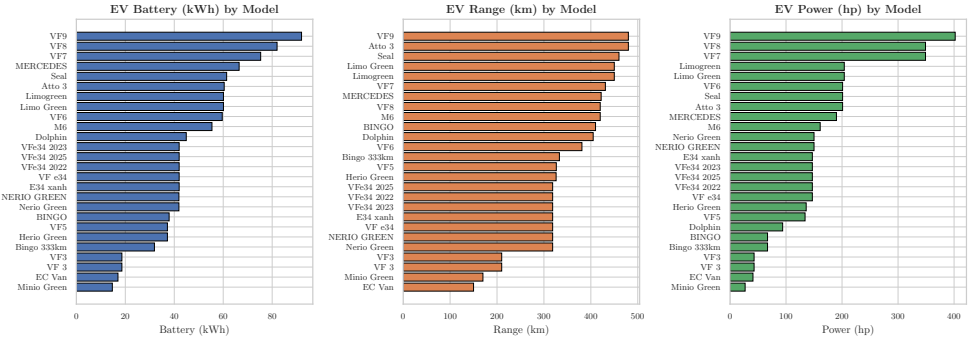


Figure: EV specifications across models — captures fundamental pricing physics

Encoding Strategy & Feature Importance

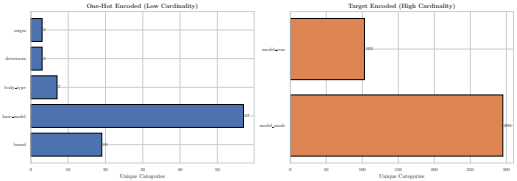


Figure: Hybrid encoding

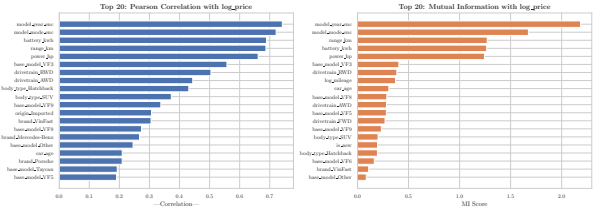


Figure: Top features by correlation & MI

Experimental Setup

Data Split:

- 80/20 train–test (2,412 / 603)
- Stratified on `base_model`
- Sample weights: inverse quintile frequency

Evaluation:

- RMSE, MAE, R^2, Adj. R^2, MAPE
- 95% CI via bootstrap (1,000 resamples)
- Segment-specific analysis

Four Models:

Model	Input
Ridge	Scaled + log target
SVR (RBF)	Scaled + log target
Random Forest	Raw features
XGBoost	Raw features

All tuned via **5-fold CV** with GridSearchCV

Overall Benchmark

Model	**RMSE** (M VND)	**MAE** (M VND)	R^2	**Adj.** R^2	**MAPE** (%)
Ridge	381	115	0.636	0.610	43.5
SVR	369	94	0.660	0.635	48.9
Random Forest	**332**	**89**	**0.725**	**0.705**	**43.6**
XGBoost	383	103	0.633	0.606	43.0

Table: Test set results ($n = 603$). RF 95% CI for RMSE: [139M, 496M] VND.

Random Forest wins across all metrics

Model Comparison

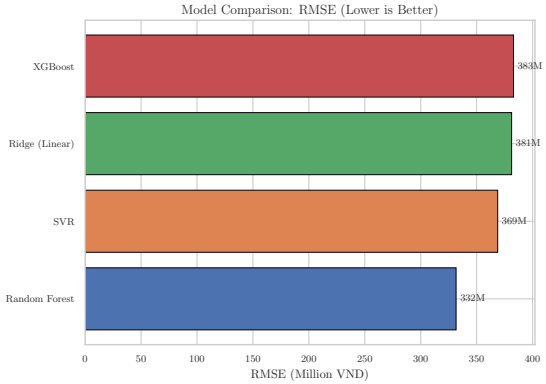


Figure: RMSE comparison

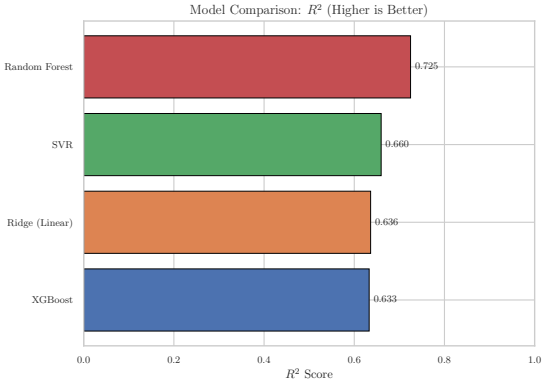


Figure: R^2 comparison

Segment-Specific Performance

Segment	RMSE	*n*
Budget (<500M)	100M	237
Mid (0.5–1.2B)	81M	318
Premium (1.2–5B)	797M	46
Luxury (>5B)	4,041M	2
Overall	**332M**	603

Table: RF by segment

Budget + Mid = **92% of test data** with practical RMSE

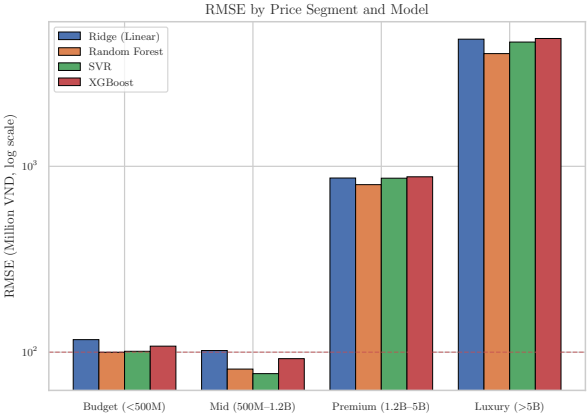
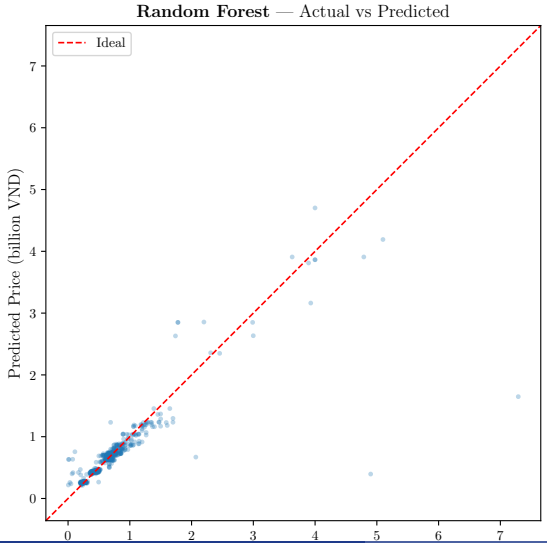
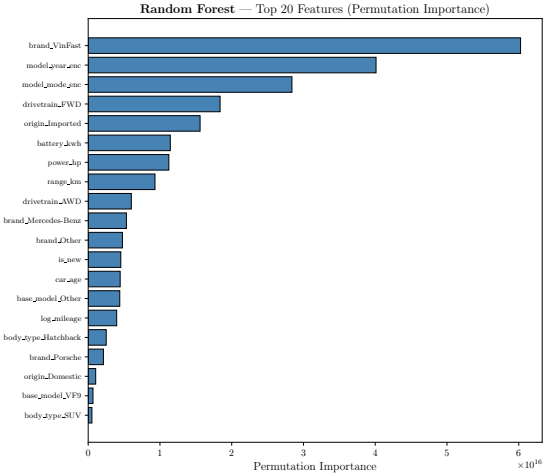


Figure: RMSE by segment (all models)

Random Forest: Predictions & Feature Importance





Data Quality vs Model Selection

Data Quality Fixes

RMSE: 1.02B → 332M VND

↓ **67% improvement**

- chotot.com price correction
- Deduplication (22% removed)
- BYD removal

Model Selection

Best vs Worst: 332M vs 383M

↓ **15% difference**

- RF vs XGBoost: 51M gap
- All models benefit equally from clean data

Takeaway

In applied ML with web-scraped data, **invest in data quality first** — it yields 4× more improvement than model experimentation.

Limitations & Future Work

Limitations:

- **Brand concentration** — VinFast = 93.6% of records
- **Small dataset** — 3,015 records (2,412 train)
- **No temporal modeling** — cross-sectional only
- **Luxury segment** — only 2 test records (n too small)

Future Work:

1. Continuous scraping $\rightarrow$ time-series depreciation model
2. Additional sources (dealer prices, registration data)
3. Ensemble stacking of all four models
4. Expand to hybrid & plug-in hybrid vehicles

Conclusion

Summary

- End-to-end pipeline: **scraping** $\rightarrow$ **LLM extraction** $\rightarrow$ **data quality** $\rightarrow$ **feature engineering** $\rightarrow$ **prediction**
- Data quality fixes reduced RMSE by **67%** (1.02B $\rightarrow$ 332M VND)
- Random Forest: $R^2 = 0.725$, RMSE $=$ 332M VND
- **Practical accuracy** for budget + mid segments (92% of test records, RMSE $\leq$ 100M)

Data quality $>$ Model selection

In real-world ML with scraped data, the dominant bottleneck is data quality, not model architecture.

Thank You

Questions?

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